Overview

Unsupervised learning discovers structure in unlabeled data. Instead of predicting known targets, models infer clusters, latent dimensions, or anomaly patterns.

Core Definitions

- **Clustering:** group samples with high intra-group similarity and low inter-group similarity.
- **Dimensionality reduction:** map high-dimensional data to lower-dimensional representation while preserving useful structure.

Clustering Methods

k-Means

Optimizes within-cluster sum of squares:

$$\min_{\{C_k\}} \sum_{k=1}^{K} \sum_{x_i \in C_k} \|x_i - \mu_k\|^2$$

Fast and popular, but assumes roughly spherical clusters.

Hierarchical Clustering

Builds cluster tree (dendrogram) via agglomerative merges or divisive splits.

DBSCAN

Density-based clustering with parameters `eps` and `min_samples`. Finds arbitrary-shaped clusters and labels noise.

Dimensionality Reduction

PCA

Linear projection maximizing explained variance.

t-SNE and UMAP (High-Level)

Non-linear methods for visualization and neighborhood preservation; useful exploratory tools, not always stable for downstream numeric decisions.

Distance and Similarity

Choice of metric (Euclidean, cosine, Manhattan) changes clustering behavior. Standardization usually required before distance-based methods.

Common Pitfalls

- Forcing k-means on non-spherical data.
- Choosing `k` only by visual preference.
- Misusing t-SNE map distances as true global geometry.

Business Case Studies & Exceptions

Case 1: Customer Segmentation

Retail team clustered purchase behavior to tailor campaigns. Segment quality degraded when seasonal behavior changed.

Mitigation: - Refit clusters periodically. - Track silhouette and segment drift metrics.

Case 2: Anomaly Detection via Clustering Distance

Using distance-to-centroid flagged anomalies, but scaling changes caused false alarms.

Mitigation: - Feature scaling pipeline lock. - Segment-aware thresholds and human review loop.

Interview Questions & Answers

1. **Q: Difference between clustering and classification?**
   A: Clustering discovers groups without labels; classification predicts predefined labels.
2. **Q: How does k-means work?**
   A: Alternate assignment to nearest centroid and centroid recomputation until convergence.
3. **Q: How choose k?**
   A: Use elbow/silhouette/domain constraints and stability checks.
4. **Q: When use DBSCAN?**
   A: When clusters are arbitrary-shaped and noise detection matters.
5. **Q: Why standardize features before clustering?**
   A: Prevent high-scale features from dominating distance calculations.
6. **Q: PCA purpose in ML workflow?**
   A: Compress features, remove redundancy, and aid visualization.
7. **Q: Limitation of t-SNE?**
   A: Great for visualization but poor for preserving global distances/cluster sizes.
8. **Q: Silhouette score interpretation?**
   A: Higher means better separation/cohesion, but not sole truth metric.
9. **Q: Unsupervised output validation challenge?**
   A: No labels, so rely on internal metrics and business usefulness tests.
10. **Q: Business risk in unsupervised segmentation?**
    A: Spurious clusters can drive bad targeting or pricing decisions.